Unit 2

Flatness & Surface Roughness

(Handwritten Notes)

Syllabus Topics Covered

1. Measurement of flatness: straight edges, surface plates
2. Optical flats, autocollimators, interferometers and their applications
3. Surface roughness terminology; roughness vs waviness
4. Numerical assessment: CLA, RMS, Ra, Rz
5. Methods of measurement: profilograph, talysurf
6. BIS symbols for indication of surface roughness

1. Flatness Measurement

A surface is **flat** when all its points lie in **one plane** . **Flatness error** = the **minimum separation of two parallel planes** between which the entire surface lies (**flatness tolerance zone**). Importance: machine-tool beds, slideways, surface plates, gauge faces, mating surfaces (leak-proof joints), optical parts.

Flatness tolerance

Surface must lie between **two parallel planes** a distance **t** apart (e.g. 0.02 mm).
Flatness controls **form** only, not orientation or position.

1.1 Straight edges

A **straight edge** is a **precision reference** with a long, **accurately flat** working edge, used to check the **straightness and flatness** of surfaces by comparing the work with the edge.

Type	Description / use
Toolmaker's (bevelled) straight edge	steel, bevelled edge ground & lapped; for small lengths (up to 300 mm); light-gap test
Wide-edge straight edge	wider face prevents tilting; checking of lathe beds, machine slides

I-section / box girder	cast iron or steel; rigid with minimum weight for lengths 1-3 m (machine tool beds)
Parallel-sided & angle straight edges	angle edge for dovetails and V-slides

- **Method 1 - Light-gap (light slit) method** : straight edge placed edge-down on the work with light behind; a **gap of 0.002 mm (2 micron)** is seen as a **white light** ; narrower gap shows blue, gap < 0.5 micron cannot be seen. Used to estimate error visually.
- **Method 2 - Engineer's blue (Prussian blue) method** : thin blue coat on the straight edge; slid over work; **high spots** show as blue patches - then scraped.
- **Method 3 - Feeler gauge / slip gauge** under edge to measure gap; **dial indicator** moved along the straight edge.
- Straight edge itself is checked by **reversal** or by comparing with a master.

1.2 Surface plates

A **surface plate** is a **flat, rigid reference plane** on which work is placed for inspection, marking-out, and as a datum for height gauges, dial indicators, sine bars, comparators and slip gauges.

- **Materials** : **cast iron** (close-grained, stress-relieved, ribbed, **scraped**), **granite** (black granite - dense, non-magnetic, **no burrs, rust-free** , low thermal expansion, hard, stable), **glass** , **ceramic** .
- **Grades (IS 2285) : Grade 00 (laboratory master) = AA , Grade 0 = A (inspection room) , Grade 1 = B (toolroom/workshop)** , Grade 2 (rough). Flatness tolerance for grade 1 (1000 x 1000 mm): ~ 10 micron.

Cast-iron plate	Granite plate
Rusts; burrs raised on scratches	Rust-free; scratch makes only a crater - no burr
Magnetic; warps with temperature	Non-magnetic; low thermal expansion
Cheaper; threaded holes & T-slots possible	Costly; brittle; chips on impact; inserts needed
Needs periodic re-scraping	Easier maintenance; wears evenly; good damping

- **Making** : cast, seasoned (aged), rough-planed, ground, then **hand-scraped** to flatness using three-plate method (Whitworth) with Prussian blue; **lapped** for

high grades. Supported at **three points (Airy points)** to avoid sagging.

- **Checking flatness of a surface plate** : (i) **Autocollimator** ; (ii) **spirit-level / precision level** on a grid; (iii) **Moody's method** with a ... (iv) **laser interferometer** .

1.3 Spirit level / precision level

- A **sensitive bubble vial** in a rigid body; the bubble moves when the base tilts. Sensitivity: **0.02 mm/m** (about 4 seconds of arc per division). Readings along **line (Moody) grid pattern** (the Union Jack pattern) give relative heights; the **plot** of heights shows flatness.

1.4 Optical flat

An **optical flat** is a **disc of fused quartz, glass or Pyrex** (diameter 25-300 mm, thickness ~ 1/6 of diameter) with **one or both faces lapped and polished** to a flatness of 0.05 micron (**1/20 of wavelength**). Used to check the **flatness of gauge faces, slip gauges, micrometer anvils** , surface plates.

- **Principle** : **interference of light** - when the flat is placed at a **small angle (wedge)** on the surface, **monochromatic light** reflected from the **bottom of the flat** and from the **work surface** interferes: **alternate dark and bright bands (fringes)** are formed.

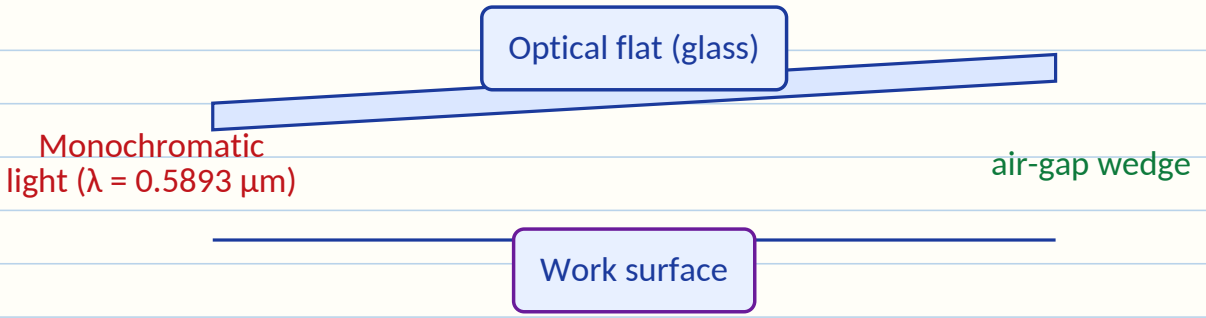


Fig 2.1: Air-wedge between optical flat and work

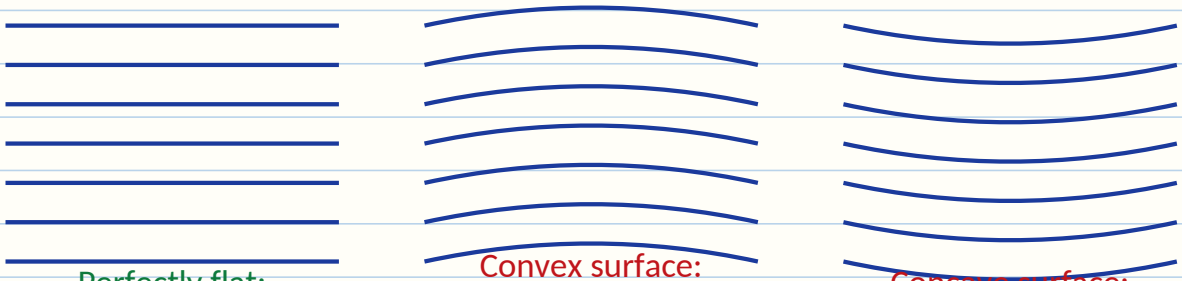
Key relation

Each **fringe** = a change in air-gap of **λ/2** .

- For monochromatic helium light λ = 0.5893 micron -> **λ/2 = 0.2945 micron (about 0.3 micron)** per fringe.
- **Flatness error = (number of fringes of curvature) x λ/2** .

- **Reason** : the extra path of the ray going through the air gap is 2t; **dark band** where 2t = nλ (with 180 degree phase change at reflection from the denser surface); **bright band** where 2t = (n + 1/2)λ.

Interpretation of fringe patterns



Perfectly flat: straight, parallel, equispaced

Convex surface: fringes curve toward the line of contact

Concave surface: fringes curve away

- **Curved fringes** : the **deviation of a fringe from straightness / spacing between fringes** = amount of departure. If a fringe bends by **a** while the **spacing is b** , **flatness error = (a / b) x λ/2** .
- **Closed circular rings (Newton's rings)** -> surface is **spherical** (convex or concave); number of rings x 0.3 micron = depth. **Saddle / cylindrical** surfaces show **parabolic bands** .
- **Wringing** a good optical flat on the gauge face -> fringes absent (full dark/ uniform) for an extremely flat surface.
- **Care** : clean, handle with soft gloves/tissue, **never slide** (scratches), store with cover; **check the flat** itself by comparing with a master flat (**three-flat method**).

2. Autocollimator

An **optical instrument** used to measure **small angular deviations** , **flatness, straightness, squareness and parallelism** . It is a combination of a **telescope and a collimator** : it projects a **parallel beam** , receives the **reflected beam** from a **mirror** (placed on the object) and measures the **tilt of the mirror** .

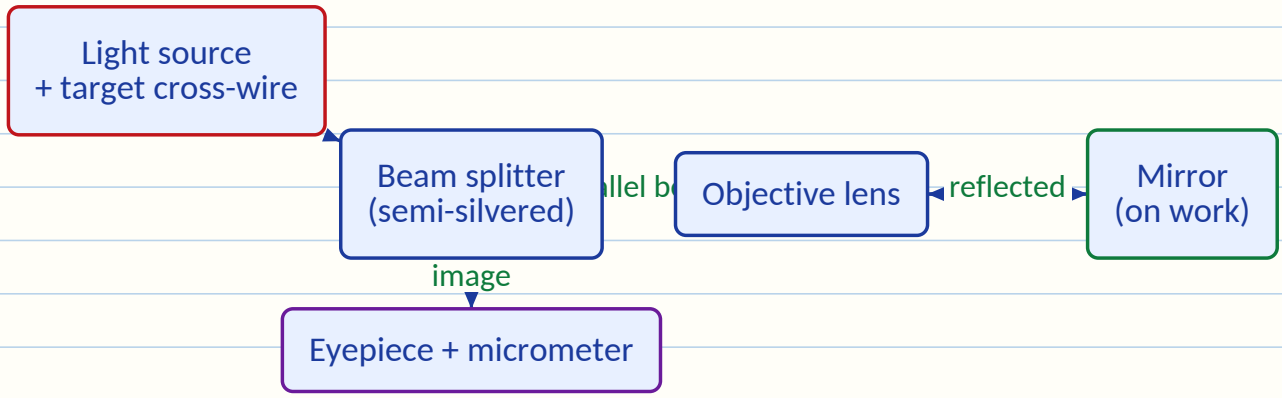


Fig 2.2: Principle of the autocollimator

- **Principle** : if the mirror is **exactly perpendicular** to the optical axis, the reflected image of the target falls on the **target itself** (coincides with the cross-wire). If the mirror **tilts by θ** , the reflected beam deviates by **2θ** , and the image is displaced by **x = 2 f θ** (f = focal length of the objective).

Basic relation

x = 2 f θ -> θ = x / (2 f)

- The displacement x is read on the **micrometer eyepiece** (1 division = 1 second of arc / 0.1 second in precise models). Range ~ 10 minutes; sensitivity 0.1 - 1 second.

- **Types** : visual (eyepiece), **digital / electronic** (CCD, photo-electric), **laser autocollimator** , **dual-axis** .

Use for flatness testing of a surface plate

1. Divide the surface into a **grid of lines** (Moody / Union Jack pattern) with the line length equal to the **base length of the mirror carriage** (e.g. 100 mm).
2. Place the **reflector (mirror) on a base** at one end; autocollimator is fixed at the other end - read the **angle** (in seconds) at each successive position.
3. Each reading gives the **tilt** of the segment; **height change** between ends of base b = **b x θ** (θ in radian: 1 second = 4.848 x 10^{-6} rad).
4. **Cumulate heights** along the line to obtain **profile** , repeat for all lines, then find the **minimum two-plane zone** -> flatness error.

- **Example** : base length 100 mm, readings 5, -3, 4 seconds -> rise = 100 x 5 x 4.848 x 10^{-6} = **0.0024 mm** for first step.
- **Applications** : flatness & straightness of **machine-tool slideways** , surface plates; **squareness** of two surfaces (with optical square / pentaprism);

parallelism ; angular measurement (along with angle dekkor); alignment of bearing bores and shaft lines ; checking small angles of tilt in machine movement (pitch, yaw).

3. Interferometers

Interferometry = the technique of using interference of light waves to measure lengths, flatness and displacement with accuracy up to a fraction of the wavelength of light . The interferometer splits a light beam into two paths, recombines them and the fringe pattern shows the path difference .

- Source requirements : monochromatic and coherent (sodium, mercury-198, cadmium, krypton-86, He-Ne laser).

3.1 NPL flatness interferometer

- Developed at National Physical Laboratory (UK) ; for calibration of slip gauges, end bars (flatness, parallelism, length) in terms of wavelength.
- Construction : mercury-vapour lamp (green light 0.5461 micron) -> condenser, slit, collimating lens -> a dispersing prism to separate the wavelengths (selects a single wavelength), then the beam -> optical flat placed over the gauge block wrung on a base plate . The reflected rays from the base plate and from the gauge block top give two sets of fringes (offset) viewed through the telescope .
- Measurement : the fringe displacement (a) relative to fringe spacing (b) gives the fractional order: length = (n + a/b) λ/2 . Using several wavelengths (red, green, blue cadmium) and the method of exact fractions the integral order n is found.

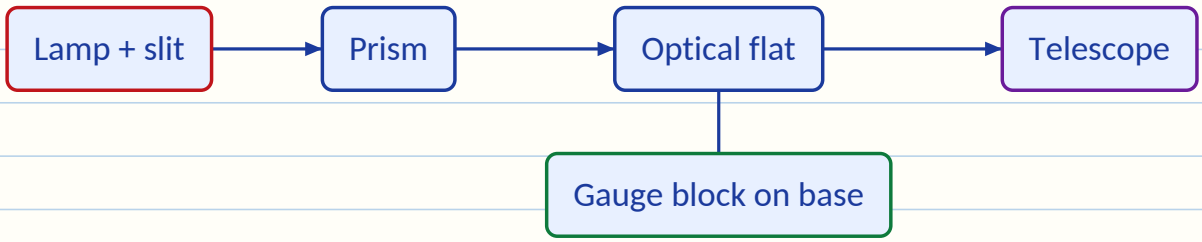


Fig 2.3: NPL flatness interferometer (simplified)

3.2 Michelson interferometer / laser interferometer

- Michelson : a beam splitter divides the beam; one goes to a fixed mirror , the other to a moving mirror (on the machine slide). When the moving mirror moves by λ/2 , the path difference changes by λ , so one fringe passes.

Counting fringes gives displacement: $d = N \lambda / 2$.

- Laser interferometer (Hewlett-Packard type): He-Ne laser (λ = 0.633 micron), stabilized ; retro-reflectors and photodetectors with reversible counters ; resolution 0.01-0.1 micron ; range up to tens of metres ; with environment compensation (air temperature, pressure, humidity).
- Applications : calibration of machine-tool positioning , CMMs , slip gauges , angular (with optics), straightness , flatness , squareness ; vibration & displacement measurement; length standard realization; optical component testing.
- Advantages : extremely high accuracy, non-contact, traceable to wavelength. Disadvantages : costly, sensitive to vibration and air disturbances , needs skilled operators.

Instrument	Measures	Typical resolution
Straight edge + feeler	straightness, flatness	0.002 mm
Spirit level	inclination, flatness	0.02 mm / m
Autocollimator	small angles, flatness	0.1 - 1 arc-second
Optical flat	flatness of small faces	0.3 micron / fringe
Laser interferometer	length, straightness, flatness	0.01 - 0.1 micron

4. Surface Roughness Measurement

No machined surface is perfectly smooth - it has irregularities left by the manufacturing process. Surface texture = repetitive or random deviations from the nominal surface - includes roughness, waviness, lay and flaws . It affects friction, wear, lubrication, fatigue strength, corrosion resistance, fits, appearance and heat transfer .

Surface profile = roughness (fine, blue) + waviness (green)

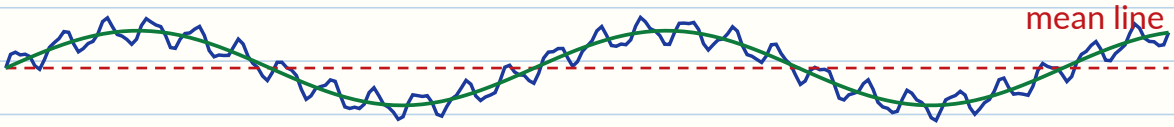


Fig 2.4: Roughness and waviness on a surface profile

Terminology (IS 3073 / ISO 4287)

Term	Meaning
Actual surface / profile	surface (trace) as it really exists - cut by a plane perpendicular to the surface
Roughness	**closely spaced** finer irregularities produced by the **cutting tool / grains of abrasive** , feed marks
Waviness	**widely spaced** irregularities (greater spacing than roughness) due to **machine vibration, chatter, deflection of work, warping, heat treatment**
Lay	direction of the **predominant surface pattern** , usually decided by the production process
Flaws	irregularities that occur **at one place / random** : scratches, cracks, blow-holes, inclusions
Sampling length (cut-off) l	length of profile used for assessing roughness: 0.08, 0.25, 0.8, 2.5, 8, 25 mm. Waviness is filtered out by the cut-off.
Evaluation length	length of profile used for assessment; normally **5 sampling lengths**
Mean line (M-system)	line so drawn that, within the sampling length, the **sum of areas above = sum of areas below** it
Peak / valley	highest point / lowest point of profile above / below the mean line
Primary profile	total profile (roughness + waviness + form) after removing nominal form

Differences between roughness and waviness

Surface roughness	Surface waviness
Closely spaced, fine irregularities	Widely spaced, longer wavelength irregularities
Pitch (spacing) **< 0.8 mm** approx.	Spacing **0.8 - 10 mm** (roughly 1000x height)
Caused by **cutting action** - tool feed, tool tip, abrasive grains, built-up edge	Caused by **vibration, chatter, machine deflection, heat treatment, clamping strain**
Height measured in **microns** ; Ra, Rz	Height measured as **Wt / Wa** (larger amplitude)
Superimposed on waviness	Superimposed on form errors

- **Typical order of magnitude of roughness: lapping 0.05-0.4 micron Ra; honing 0.1-0.8; grinding 0.1-1.6; reaming 0.4-3.2; turning/milling 0.8-6.3; shaping/planing 1.6-12.5; sand casting 12.5-25; forging 3.2-12.5** .

5. Numerical Assessment of Surface Finish

Surface finish is numerically assessed by **parameters** computed from the profile **within the sampling length L** relative to the **mean line** ; y = height of the profile above (+) / below (-) the mean line at distance x.

1. Ra - Centre Line Average (CLA) / Arithmetic average

Ra = (1/L) $\int_0^L |y|\,dx$ (approx. **Ra = (|y1| + |y2| + ... + |yn|) / n**)

- Ra = **arithmetic mean of absolute deviations** of the profile from the mean line. Most widely used; unit **micron (µm)** .
- Also called **AA (arithmetic average)** or **CLA (centre line average, British)** .

2. Rq - RMS value (root mean square)

Rq = sqrt[(1/L) $\int_0^L y^2\,dx$] ≈ sqrt[$(y1^2 + y2^2 + ... + yn^2) / n$]

- More sensitive to peaks and valleys than Ra. **For a sine wave: Rq = 1.11 Ra** (approx. relation **RMS = 1.11 CLA**). Used in USA (RMS in micro-inch).

3. Rz - Ten-point height of irregularities

Rz = [(R1 + R3 + R5 + R7 + R9) - (R2 + R4 + R6 + R8 + R10)] / 5

- (average of the **five highest peaks** minus average of the **five deepest valleys**) in the sampling length, measured from a line parallel to the mean line. (DIN Rz = mean of 5 successive Rmax.)

Parameter	Name / description
Rmax (Ry, Rt)	**maximum peak-to-valley height** within the sampling (evaluation) length
Rp / Rv	maximum **peak height** above / maximum **valley depth** below the mean line
Rsm	mean **spacing** of profile elements (peaks) - lateral parameter
Rmr (tp)	**material ratio** (bearing ratio) at a given depth - Abbott-Firestone curve; predicts wear-in & load carrying

Example : profile ordinates (micron): +3, -2, +4, -5, +2, -4 -> Ra = (3 + 2 + 4 + 5 + 2 + 4)/6 = **3.33 micron** ; Rq = sqrt[(9 + 4 + 16 + 25 + 4 + 16)/6] = sqrt(12.33) = **3.51 micron** .

- **Why Ra alone is insufficient** : two different profiles (sharp peaks vs. wide plateaus) can have the same Ra - hence Rz, Rmax, Rsm and Rmr are used with it.
- **Roughness grade numbers (N-series, ISO 1302)** : N1 = 0.025 micron Ra, N2 = 0.05, N3 = 0.1, N4 = 0.2, N5 = 0.4, N6 = 0.8, N7 = 1.6, N8 = 3.2, N9 = 6.3, N10 = 12.5, N11 = 25, N12 = 50 micron.

6. Methods of Measurement of Surface Finish

Method	Examples	Remarks
Comparison (qualitative)	touch, visual, **fingernail** , **surface-roughness comparison specimens** (standard plates for turning, milling, grinding), scratch/viewing with microscope	rough, subjective, quick - shop floor
Direct (stylus)	**Talysurf** , profilometer, **profilograph** (surface analyzer)	quantitative; Ra, Rz, profile record

Optical	light-section microscope (Schmaltz) , interference microscope (Tolansky) , double-slit, speckle	non-contact; for very smooth surfaces
Other	pneumatic, capacitance, scanning probe (AFM, SEM)	nano scale

6.1 Stylus (Talysurf / profilograph) method

A diamond stylus (tip radius 2-10 micron, 90 degree cone) is traversed across the surface ; the vertical motion of the stylus (caused by roughness) is converted by a transducer into electrical signals, amplified, filtered and given as a profile (graph) or Ra reading .

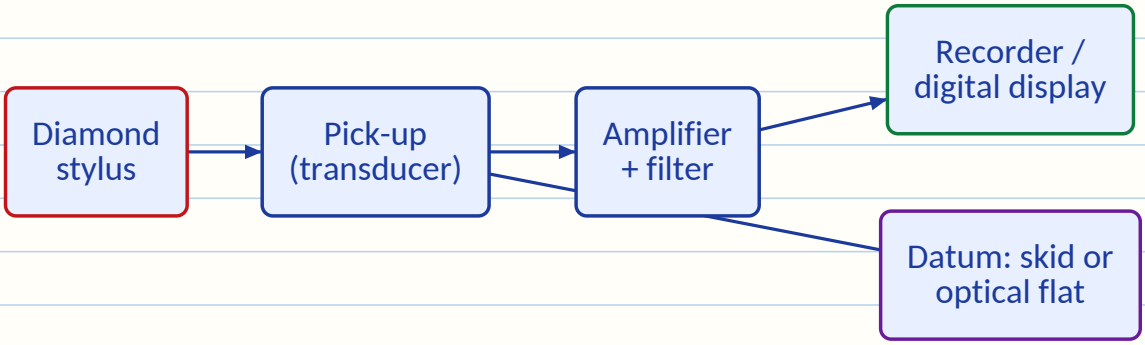


Fig 2.5: Block diagram of Talysurf

- Talysurf (Taylor-Hobson): the stylus is attached to an arm pivoted on a knife edge ; the other end carries an armature between two E-shaped coils (inductive pick-up, differential). Roughness moves the armature -> unbalance of the AC bridge -> modulated signal -> demodulated, amplified -> recorded; Ra computed electronically.
- Datum : skid (a blunt, long-radius shoe riding on the surface; the skid follows the waviness - acts as a mechanical filter), or an independent datum (optical flat / precision slide) when waviness has to be measured.
- Drive unit : motor driving the pick-up at a constant traverse speed (e.g. 1 mm/s); traverse length = evaluation length; cut-off filter (0.25, 0.8, 2.5 mm) (electrical 2RC filter / Gaussian).
- Magnification : vertical 500 - 100,000x; horizontal 10 - 1000x. Measuring range 0.01 - 100 micron.
- Limitations : stylus radius can't enter very fine valleys; stylus pressure may

scratch soft surfaces; vibration of the shop; slow .

6.2 Profilograph / Profilometer

- **Profilograph** (Surface analyzer): a **recording** instrument that **draws the magnified profile** (like a chart) of the surface; the stylus movement is amplified electronically and **plotted** on paper.
- **Profilometer** : a **meter-type** instrument (Brush Surfindicator, Perthometer) - the stylus **traces** the surface and a **pointer** (or digital display) shows **Ra** or RMS **directly** , without graph; **fast** , convenient on the shop floor and **portable** .

6.3 Optical methods

- **Light-section (Schmaltz) microscope** : a thin **slit of light** at **45 degree** to the surface forms a **bright line image** that follows the profile; viewed through the microscope at **90 degree** - magnified profile (0.8-80 micron).
- **Interference microscope (Tolansky)** : optical flat over the surface; **fringes** displaced by the irregularities - **fringe displacement = λ/2** per fringe; for **very fine finishes** (< 0.025 micron).

7. BIS Symbols for Indication of Surface Roughness

IS 696 / IS 3073 / IS 10719 (ISO 1302) : standardized symbols on **engineering drawings** to specify the **surface finish** and **machining** requirements.

Symbol	Meaning
Basic symbol (two unequal legs, 60 degree)	surface **texture may be obtained by any method** ; alone it has no meaning without further information
Basic symbol + **horizontal bar** (closed triangle)	**machining required** (material removal necessary - cutting, grinding, etc.)
Basic symbol + **circle**	**material removal prohibited** - surface left as from casting, forging, rolling, etc.
Symbol + **circle on the closed line**	same surface finish required on **all surfaces** around the contour

Position of specifications on the symbol:

a = Ra value (µm) or N grade	b = production method, treatment, coating	c = sampling length (cut-off, mm)
d = lay direction symb	e = machining allowance (mm)	f = other roughness (Rz)

Lay symbols

Symbol	Direction of lay	Example
=	**parallel** to the plane of projection of the view	shaping, planing
⊥	**perpendicular** to the plane of projection	shaping (ends), turned face
X	**crossed** in two slant directions	honing
M	**multi-directional** (no particular direction)	lapping, grinding, super-finishing
C	approximately **circular** relative to the centre	face turning
R	approximately **radial** relative to the centre	end milling

Indication on drawings

- The symbol is drawn **on the outline of the surface or on an extension line** , **pointing to the surface from outside the material** (like a dimension) and normally **read from the bottom or right** side of the drawing.
- Example: **Ra 1.6** (or **N7**) written above the symbol = roughness average **1.6 micron** ; **max** and **min** values can be given (e.g., Ra 3.2 max, Ra 1.6 min). The sampling length is given below the horizontal bar (e.g., **0.8**); machining allowance **2 mm** .
- **Common finishes / uses** : **N12-N10** rough surfaces (non-mating); **N9-N8** medium (general machining); **N7-N6** fine (mating parts, bearings seats); **N5-N4** very fine (piston, journals, **slideways**); **N3-N1** super-finish (gauges, lapped, **slip gauges**).

Quick Revision / Important Questions

1. Flatness: straight edges, surface plates, optical flat - principle and fringe patterns.

2. Explain the autocollimator principle and its use for flatness.
3. Interferometers (NPL, laser) and their applications.
4. Roughness vs waviness; Ra, RMS, Rz formulas with examples.
5. Talysurf and profilograph; BIS symbols and lay.